

Code-Layout, Readability and Reusability

Date	2 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Python follows PEP 8 indentation standards.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for dataset, model, Flask app, templates, static files, and notebooks.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables like applicant_income, credit_history, prediction, and model are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions such as train_model(), preprocess_data(), predict_approval() clearly indicate their purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Comments are added for major sections, but more inline comments can improve readability.
6	Modular Design	Code is split into reusable functions/modules	Yes	Data preprocessing, model training, prediction, and Flask routes are implemented as separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common preprocessing and prediction logic are reused across the project.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation and Flask error handling are implemented; more exception handling can be added.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Preprocessing Module	Python, Pandas, NumPy	Training and prediction pipeline	High
2	Machine Learning Model Training	Scikit-Learn, XGBoost	Model development and evaluation	High
3	Prediction Module	Python, Pickle	Flask application and IBM Watson deployment	High
4	Flask Web Application	Flask (Python)	User interface for credit approval prediction	Medium
5	IBM Watson ML Deployment	IBM Watson Machine Learning	Cloud-based real-time prediction service	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with clear folder hierarchy and modular implementation.
Readability	4	Meaningful naming conventions improve readability; additional comments can further enhance understanding.
Reusability	5	Preprocessing, model training, and prediction modules are reusable across different workflows.
Documentation / Comments	4	Documentation is available, but more inline comments would improve maintainability.
Overall Score	4.5	High-quality, modular machine learning project suitable for future enhancements and deployment.